

Sahayak

A voice-first AI work assistant for India's 400 million deskless workers — on WhatsApp, in their own language.

🗣️ “Meri kitni leave bachi hai?”

Concept-to-Prototype Execution Plan

Harshil Patel · Deep Mehta · Meet Kapadia · IIIT Vadodara

Enterprise software forgot 90% of India's workforce

400M+

deskless workers — nurses, operators, pickers, technicians, retail staff

~10%

of Indians speak English, yet nearly all enterprise tools are English-only

40–60%

of HR tickets are repetitive questions answered by humans, over and over

What that exclusion looks like every day:

- 1 A ward nurse can't check her shift roster or leave balance — she queues at the HR desk instead.
- 2 A machine operator hits "Error 42"; the fix exists in an English PDF nobody can search. The line stays down.
- 3 Workers miss PF/ESI entitlements they never knew they had; safety SOPs go unread until an incident.

Sahayak: send a voice note, get work done

Sahayak · online

🗣️ “Meri kitni leave bachi hai?”

Aapke paas 6 casual leave bachi hain (HRMS, aaj).

🗣️ “Kal ki chhutti apply kar do.”

Ho gaya ✓ Leave #4821 filed for 21 Jul. Supervisor ko notification chala gaya.

🗣️ “Machine X error 42?”

SOP p.12: coolant valve reset — steps + photo attached 📷

1

Answers

From the employer’s own systems — HRMS, rosters, payslips — and SOP manuals via RAG, with the source page cited.

2

Actions

Files the leave, raises the ticket, requests the shift swap — with explicit confirmation and a full audit log.

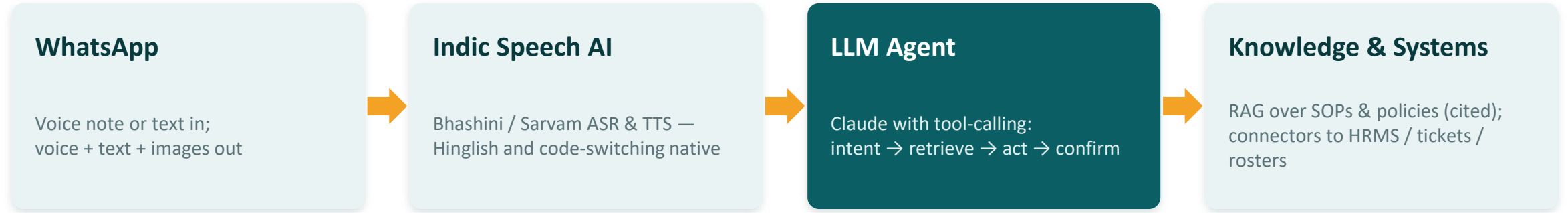
3

Insight

Anonymized query clusters become a live training-gap heatmap for HR and supervisors.

Zero install · zero training · voice-first in Hindi, Hinglish & regional languages

How it works



Guardrails at every step: answers cite source document & page · low confidence → “I don’t know” + human handoff · every action needs explicit confirmation · full audit log

Supervisor dashboard (web):

live query clusters, training-gap heatmap, action log, adoption metrics — the employer’s window into what the workforce needs.

What judges will see: a 3-minute live demo

1

Ask by voice

A Hindi voice note on a real phone:
“Infection control ka hand-hygiene SOP kya hai?”

Nurse persona, hospital corpus

2

Cited answer

Sahayak replies in Hindi — voice + text —
citing the SOP document and page

RAG, not hallucination

3

Take action

“Kal ki leave apply kar do” → confirmation →
leave filed in the HRMS

Agentic, closed-loop

4

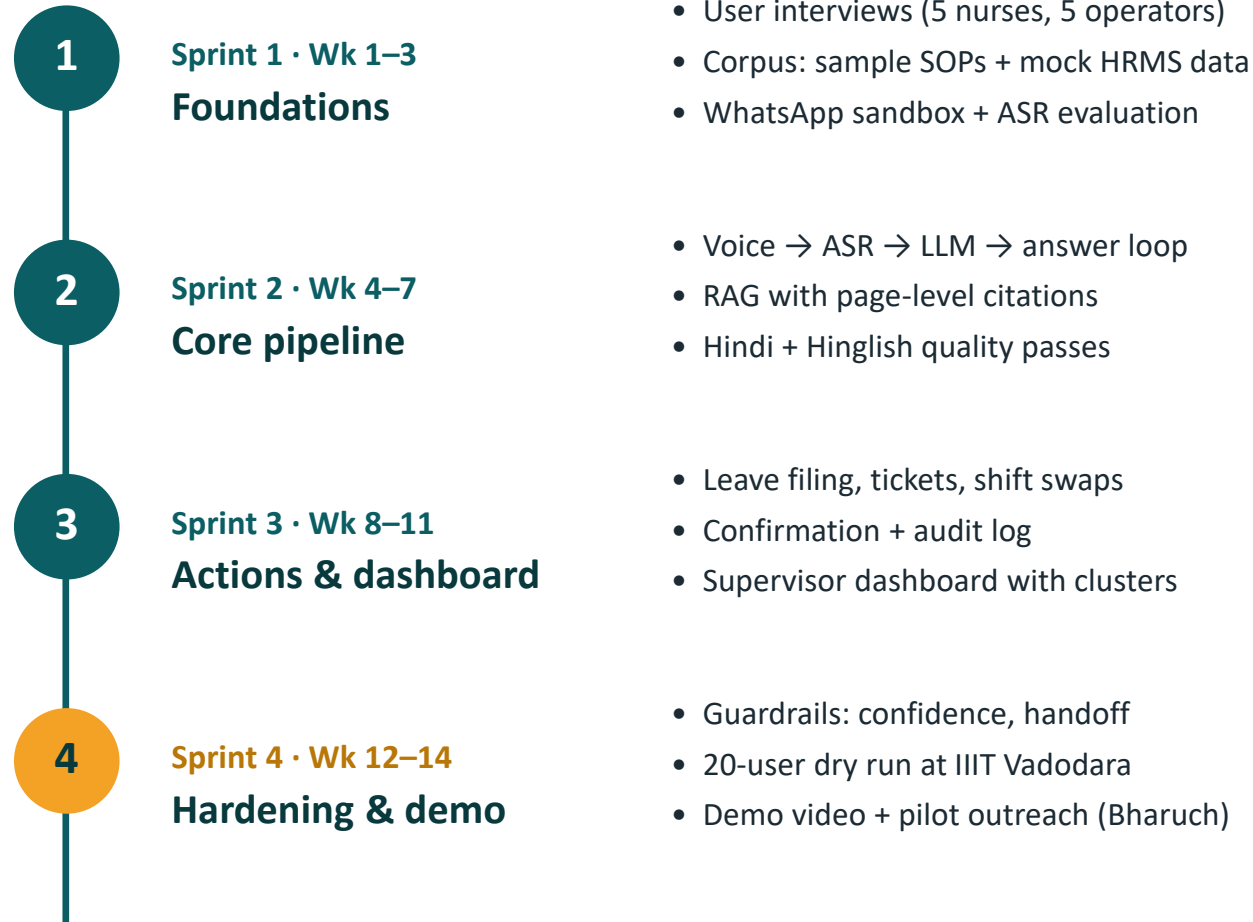
See the insight

The HR dashboard updates live: query logged,
cluster grows, training gap surfaces

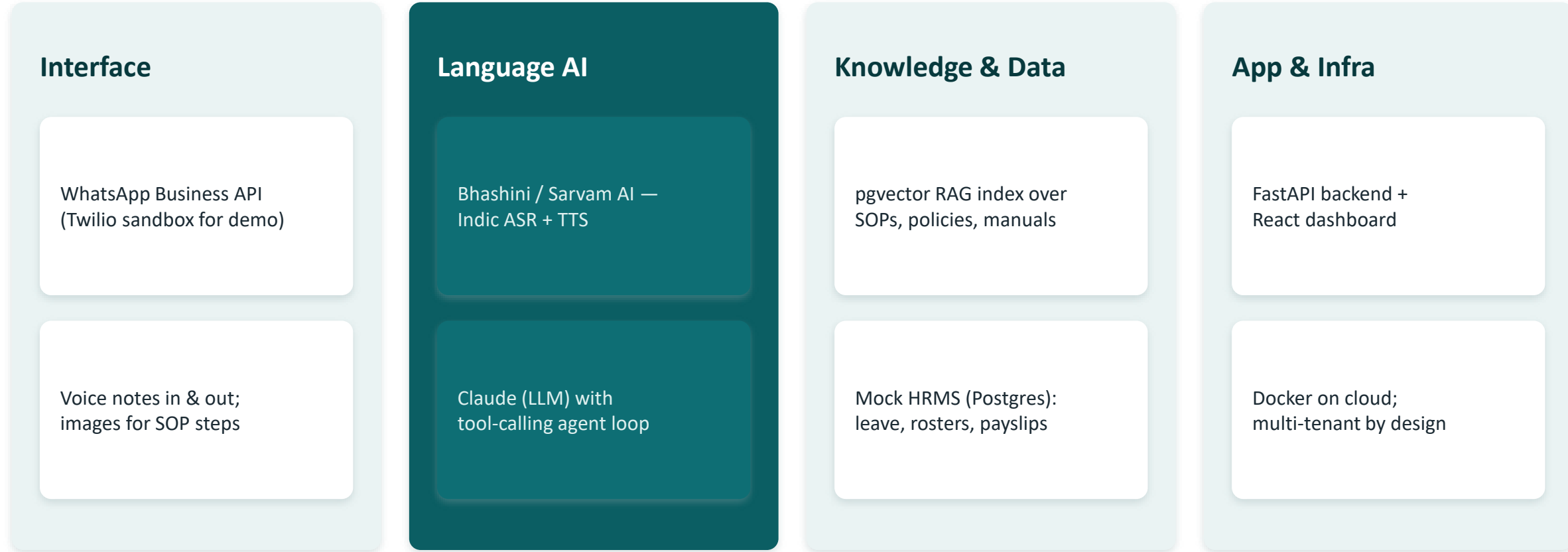
Value for the employer

Concept → prototype: 14 weeks, four sprints

Jul 1 – Oct 7, 2026 (challenge window)



Tech stack: proven parts, novel integration



Every component is individually proven — our work is the integration, the Indic voice UX, and the action layer.

How we'll know it works

$\geq 70\%$

of worker queries resolved
at first contact

30–40%

repetitive HR/IT workload
deflected from humans

$\geq 60\%$

monthly active adoption
among pilot workers

$\geq 90\%$

of answers with correct
source citation

Validation ladder

1 Hackathon (now)

Working prototype: live voice → cited answer → filed action, measured on a 200-question Hindi/Hinglish test set.

2 Dry run (Wk 12–14)

20 students + staff at IIIT Vadodara use Sahayak for a week; we measure resolution, latency, and voice-recognition accuracy.

3 Pilot (post-challenge)

6-month structured pilot at one hospital and one manufacturing SME near Bharuch, Gujarat — metrics above, plus entitlement uptake.

Inclusive by mechanic, trustworthy by design

Inclusion — built in, not bolted on

- Voice-first, both directions — literacy is never a barrier
- Hindi, Hinglish & regional languages, code-switching as spoken
- Runs on WhatsApp on entry-level phones, low bandwidth
- Zero training: if you can send a voice note, you can use it
- Same product for a nurse in Coimbatore and a welder in Pune

Trust & safety guardrails

- Every answer cites its source document and page
- Low confidence → “I don’t know” + route to a human
- Explicit confirmation before any HR action; full audit log
- Consent-first onboarding, data minimization, DPDP-aligned
- Anonymized, aggregated analytics only — never surveillance

A real market, a clear wedge

Business model

B2B SaaS at ₹30–50 per employee per month.

A 500-worker SME pays ~₹15–25k/month — less than one clerk's salary for a 24×7 multilingual helpdesk.

Verticals: hospitals → factories → warehouses → retail.

Honest competitive view

Adjacent players exist — Hunar.ai (frontline hiring & engagement), HROne One AI (HRMS voice agent), Vahan (blue-collar recruiting). None combine SOP troubleshooting RAG + agentic HR actions + training-gap analytics in one worker-side assistant.

Our wedge

1

Worker-side, not HR-side

Built for the deskless worker's daily reality, not the HR office

2

SOP brain

Shop-floor troubleshooting from the employer's own manuals, cited

3

Learning loop

Query clusters become a training-gap heatmap — a loop no competitor offers

4

Healthcare-first

Hospital pilot aligned with the challenge's health pillar

The team

HP

Harshil Patel

Product & AI pipeline — LLM agent, RAG, prompt & tool design

DM

Deep Mehta

Backend & integrations — WhatsApp, HRMS connectors, infra

MK

Meet Kapadia

Voice UX & dashboard — Indic ASR/TTS tuning, React frontend

B.Tech students, IIIT Vadodara · based in Gujarat — pilot outreach in Bharuch

***UPI made payments work for every Indian.
Sahayak does the same for the workplace.***